Lab 3: Architectural Pattern Selection — Written Justification

Pet Adoption & Shelter Management System (Problem Statement #57)

Architecture Selection:

We chose Layered Architecture for the Pet Adoption & Shelter Management System. The system is organized into a Presentation Layer, Business Layer, and Data Layer, with the main system functions separated from data storage.

Reason 1 — Clear Separation of Distinct System Functions

The system supports different functions such as pet search and filtering, adoption application processing, animal medical and vaccination management, and foster placement management. The Presentation Layer contains the Pet Portal and Staff Portal, while the Business Layer contains the main system components. The Data Layer contains the Pet Adoption Database. This separation makes the system easier to understand, develop, test, and maintain because each layer has a specific responsibility.

Reason 2 — Suitable for the Project Size and Maintainability

The system has two main types of users: adoption applicants and shelter staff, with a moderate workload. Layered Architecture is suitable for this project because it provides clear separation without the additional complexity of Microservices, such as managing multiple independent services and their communication. It also provides more structure than a simple Client-Server design by separating the system's business functions from database access. This makes the system easier to modify and extend in the future.

Security Advantage

The layered structure helps protect sensitive information by keeping database access within the Database / Data Access Layer instead of allowing the user interfaces to directly access the database. Access control and validation can be applied before information reaches the database. This helps protect sensitive data such as animal medical and vaccination records and applicant information from unauthorized access.

Performance Benefit

The Database / Data Access Layer provides a central place to manage database queries. Frequently used operations such as pet search and filtering can be optimized at this layer without requiring major changes to the presentation or business components. This can improve response time while keeping the rest of the system organized and easier to maintain.

Components shown in the UML Component Diagram:

Pet Portal, Staff Portal, Pet Search & Filter, Adoption Application, Animal Health Management, Foster Placement, Database / Data Access Layer, and Pet Adoption DB. The components are connected using appropriate <<use>> dependencies and provided/required interfaces.